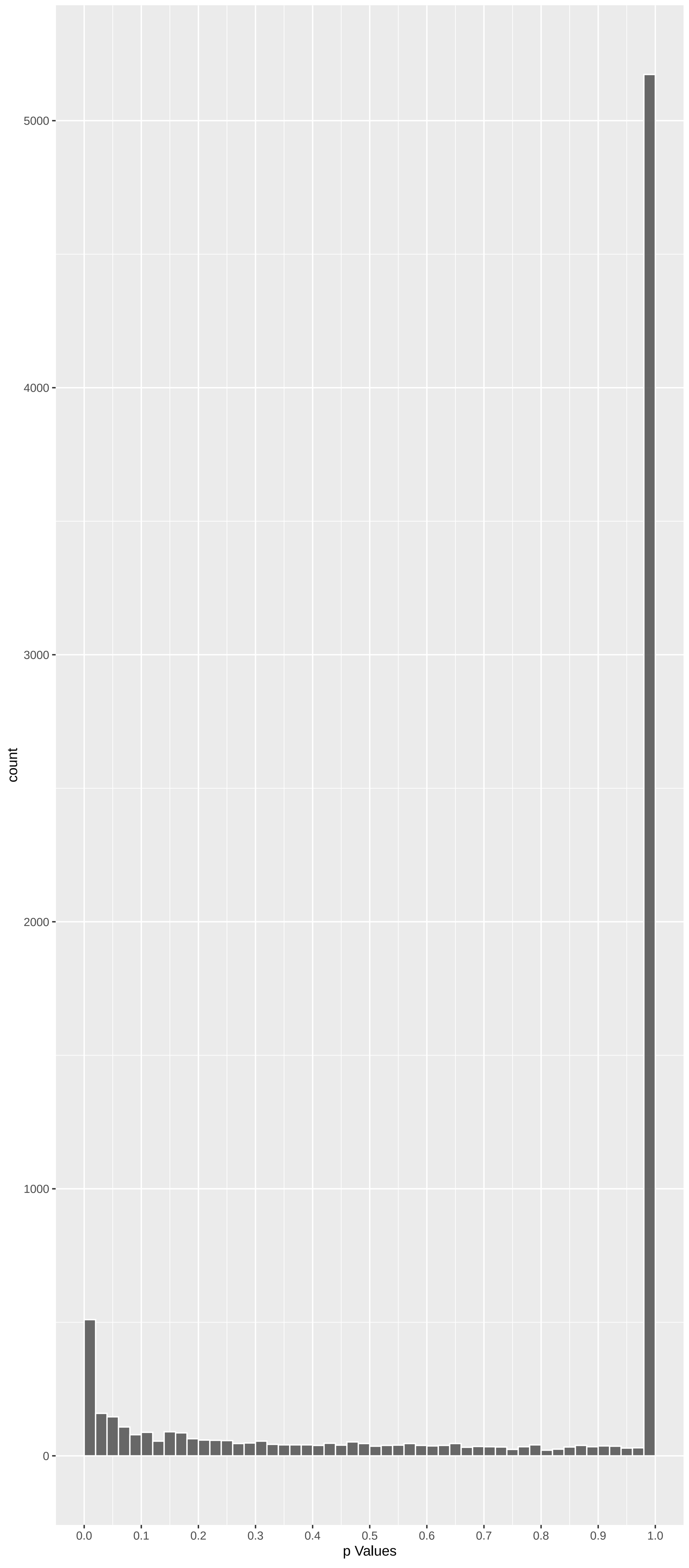
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
KIAA0753	−7.336815	1.312422e−12	2.096e−08	3.346e−04
PGBD1	−6.900036	3.119364e−11	1.245e−07	4.970e−04
TRIM15	−6.904090	3.031565e−11	1.245e−07	4.970e−04
DNAH14	−6.995640	1.584309e−11	1.245e−07	4.970e−04
PKD1L1	−6.734952	9.839033e−11	3.142e−07	1.003e−03
CEMIP	−6.455311	6.479824e−10	1.662e−06	2.654e−03
MINAR1	−6.427548	7.780696e−10	1.662e−06	2.654e−03
CAPN8	−6.383117	1.041114e−09	1.662e−06	2.654e−03
IQANK1	−6.395333	9.611885e−10	1.662e−06	2.654e−03
DNAH2	−6.394587	9.658906e−10	1.662e−06	2.654e−03
PIEZO1	−6.326665	1.503102e−09	2.043e−06	2.718e−03
PROM1	6.323373	1.535483e−09	2.043e−06	2.718e−03
PLEC	−6.257898	2.341201e−09	2.876e−06	3.532e−03
FBXW10	−6.224272	2.902777e−09	3.311e−06	3.776e−03
SLC28A1	−6.160457	4.352131e−09	4.633e−06	4.931e−03
ZBED9	−6.107704	6.064493e−09	6.052e−06	6.040e−03
POM121L2	−5.964882	1.468865e−08	1.338e−05	1.187e−02
PINLYP	5.960518	1.508637e−08	1.338e−05	1.187e−02
SFSWAP	−5.917140	1.965530e−08	1.652e−05	1.388e−02
RGPD8	5.859828	2.780073e−08	2.219e−05	1.772e−02
MYH13	−5.772299	4.691828e−08	3.567e−05	2.517e−02
ABCC11	5.764264	4.920900e−08	3.571e−05	2.517e−02
B3GNT6	−5.754234	5.222125e−08	3.625e−05	2.517e−02
SH2D7	−5.722504	6.297916e−08	4.190e−05	2.788e−02
MTBP	5.684484	7.872498e−08	5.028e−05	2.934e−02

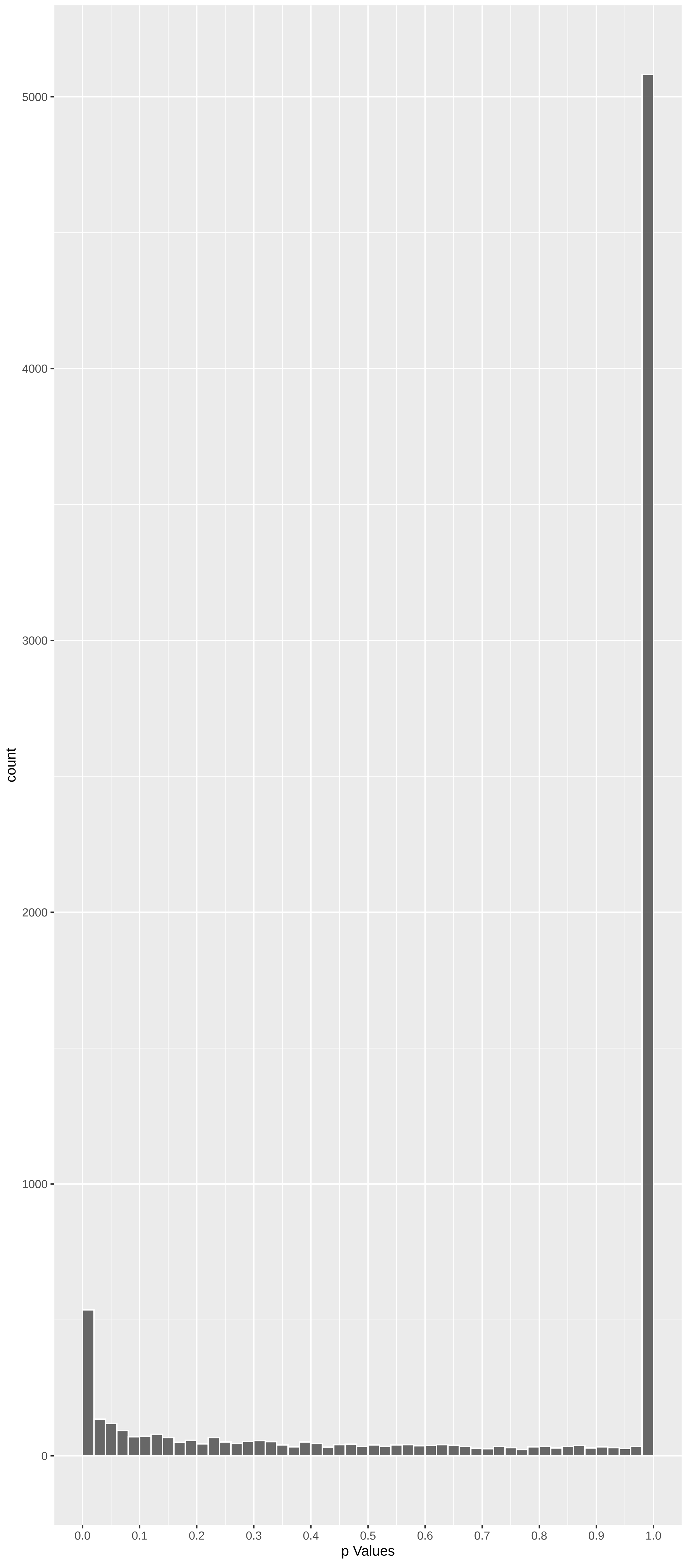
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
KIAA0753	−7.336815	1.312422e−12	2.096e−08	3.346e−04
PGBD1	−6.900036	3.119364e−11	1.245e−07	4.970e−04
TRIM15	−6.904090	3.031565e−11	1.245e−07	4.970e−04
DNAH14	−6.995640	1.584309e−11	1.245e−07	4.970e−04
PKD1L1	−6.734952	9.839033e−11	3.142e−07	1.003e−03
CEMIP	−6.455311	6.479824e−10	1.662e−06	2.654e−03
MINAR1	−6.427548	7.780696e−10	1.662e−06	2.654e−03
CAPN8	−6.383117	1.041114e−09	1.662e−06	2.654e−03
IQANK1	−6.395333	9.611885e−10	1.662e−06	2.654e−03
DNAH2	−6.394587	9.658906e−10	1.662e−06	2.654e−03
PIEZO1	−6.326665	1.503102e−09	2.043e−06	2.718e−03
PROM1	6.323373	1.535483e−09	2.043e−06	2.718e−03
PLEC	−6.257898	2.341201e−09	2.876e−06	3.532e−03
FBXW10	−6.224272	2.902777e−09	3.311e−06	3.776e−03
SLC28A1	−6.160457	4.352131e−09	4.633e−06	4.931e−03
ZBED9	−6.107704	6.064493e−09	6.052e−06	6.040e−03
POM121L2	−5.964882	1.468865e−08	1.338e−05	1.187e−02
PINLYP	5.960518	1.508637e−08	1.338e−05	1.187e−02
SFSWAP	−5.917140	1.965530e−08	1.652e−05	1.388e−02
RGPD8	5.859828	2.780073e−08	2.219e−05	1.772e−02
ABCC11	5.764264	4.920900e−08	3.571e−05	2.517e−02
MYH13	−5.772299	4.691828e−08	3.567e−05	2.517e−02
B3GNT6	−5.754234	5.222125e−08	3.625e−05	2.517e−02
SH2D7	−5.722504	6.297916e−08	4.190e−05	2.788e−02
HIRA	5.661116	9.023490e−08	5.146e−05	2.934e−02

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
PROM1	6.323373	1.535483e−09	2.043e−06	2.718e−03
PINLYP	5.960518	1.508637e−08	1.338e−05	1.187e−02
RGPD8	5.859828	2.780073e−08	2.219e−05	1.772e−02
ABCC11	5.764264	4.920900e−08	3.571e−05	2.517e−02
MTBP	5.684484	7.872498e−08	5.028e−05	2.934e−02
TMEM131L	5.676282	8.259225e−08	5.072e−05	2.934e−02
HIRA	5.661116	9.023490e−08	5.146e−05	2.934e−02
MGA	5.577249	1.466109e−07	7.315e−05	3.650e−02
NAA16	5.550158	1.712468e−07	8.286e−05	4.009e−02
LRP4	5.407898	3.826121e−07	1.608e−04	6.566e−02

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
KIAA0753	−7.336815	1.312422e−12	2.096e−08	3.346e−04
PGBD1	−6.900036	3.119364e−11	1.245e−07	4.970e−04
TRIM15	−6.904090	3.031565e−11	1.245e−07	4.970e−04
DNAH14	−6.995640	1.584309e−11	1.245e−07	4.970e−04
PKD1L1	−6.734952	9.839033e−11	3.142e−07	1.003e−03
CEMIP	−6.455311	6.479824e−10	1.662e−06	2.654e−03
MINAR1	−6.427548	7.780696e−10	1.662e−06	2.654e−03
CAPN8	−6.383117	1.041114e−09	1.662e−06	2.654e−03
IQANK1	−6.395333	9.611885e−10	1.662e−06	2.654e−03
DNAH2	−6.394587	9.658906e−10	1.662e−06	2.654e−03

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Alpha−Amino Acid Biosynthetic Process (G	0.26902178	10	3.222e−03	9.205e−01	CAD:346 BCAT2:565 SDS:598 SDSL:604 MTHFD1:847 ILVBL:1549
Bile Acid And Bile Salt Transport (GO:00	0.20679031	16	4.191e−03	9.205e−01	ABCC11:4 SLC10A6:86 SLCO1A2:455 SLC27A5:1052 SLCO1B1:1485 SLC10A2:1495
Cellular Response To Acetylcholine (GO:1	−0.30102484	10	9.804e−04	9.205e−01	CHRNB2:60 CHRM3:241 CHRNA3:347 LY6H:375 CHRND:603 LYPD1:2204
Chromatin Remodeling (GO:0006338)	0.06047363	200	3.297e−03	9.205e−01	HIRA:7 CHD8:27 GATAD2A:43 DAXX:80 RNF168:116 DDX21:145
Cilium−Dependent Cell Motility (GO:00602	−0.20843465	17	2.931e−03	9.205e−01	DNAH2:9 TEKT1:157 DNAH17:186 TEKT2:1023 GAS8:1325 TEKT4:1573
Glycerophospholipid Metabolic Process (G	0.12274174	56	1.501e−03	9.205e−01	PLA2G2D:205 ABHD3:368 PLA2G7:432 PIGG:586 SMPD4:707 PIGV:769
Mitochondrial Cytochrome C Oxidase Assem	−0.21443926	15	4.038e−03	9.205e−01	FASTKD3:648 TIMM21:735 COA1:1637 COX16:1767 SURF1:1958 COA8:2061
Muscle Contraction (GO:0006936)	−0.09908076	87	1.422e−03	9.205e−01	MYH13:18 TACR2:22 CHRNB2:60 ARHGEF11:147 MYH8:352 CHRNB1:383
Negative Regulation Of CD8−positive, Alp	−0.34738851	6	3.210e−03	9.205e−01	ZBTB7B:227 SOCS1:511 HFE:1822 VSIR:1946 SLC4A2:2454 DAPL1:7900
Neuropeptide Signaling Pathway (GO:00072	−0.11308371	63	1.927e−03	9.205e−01	NPBWR2:75 SSTR4:256 CYSLTR2:353 OPRK1:396 RXFP3:397 GALR1:440
Nuclear Pore Organization (GO:0006999)	0.21692473	14	4.955e−03	9.205e−01	NUP98:502 NDC1:514 NUP133:946 NUP107:1042 NUP205:1069 NUP54:1249
Positive Regulation Of Cell Cycle Proces	0.08618947	105	2.316e−03	9.205e−01	KIF14:54 ZNF16:125 EGF:181 KIF20B:276 STIL:291 SMC2:311
Positive Regulation Of Chromosome Segreg	0.21401943	17	2.254e−03	9.205e−01	SMC2:311 NUMA1:606 CDC6:885 NCAPH:1298 RCC2:1413 SMC4:1577
Positive Regulation Of Histone Deacetyla	0.26303025	12	1.607e−03	9.205e−01	SMARCA5:399 PRKD2:1177 VEGFA:1244 SIRT1:1319 BAZ2A:1376 SIRT6:1400
Positive Regulation Of Protein Deacetyla	0.24504001	13	2.222e−03	9.205e−01	SMARCA5:399 FNTA:887 IFNG:1099 SIRT1:1319 BAZ2A:1376 NIPBL:2199
Positive Regulation Of Transcription By	0.29906611	12	3.346e−04	9.205e−01	CHD8:27 DDX21:145 ERCC6:204 SMARCA5:399 ELL:471 ZC3H8:725
Positive Regulation Of Tyrosine Phosphor	0.11929692	49	3.890e−03	9.205e−01	TNF:259 IL23R:345 IGF1:669 PARP9:686 CNTF:754 CSH2:789
Preassembly Of GPI Anchor In ER Membrane	0.30978785	7	4.535e−03	9.205e−01	PIGG:586 PIGV:769 PIGB:788 PIGN:932 PIGW:2007 PIGL:8068
Regulation Of Rho Protein Signal Transdu	−0.10368282	68	3.141e−03	9.205e−01	EPS8L3:57 RIPOR2:104 KANK1:117 ARHGEF2:221 GPR35:441 KANK2:601
Regulation Of Cell Cycle (GO:0051726)	−0.05211947	260	3.970e−03	9.205e−01	BIRC3:45 RNF167:132 BIRC7:136 TBRG4:169 FIGNL1:292 ZBTB17:380
Regulation Of Chromosome Separation (GO:	0.27222758	9	4.685e−03	9.205e−01	SMC2:311 NUMA1:606 NCAPH:1298 SMC4:1577 NCAPG2:1675 NCAPD2:2960
Regulation Of Interleukin−13 Production	0.27427262	9	4.383e−03	9.205e−01	IRF4:529 SPHK2:530 IL4:952 TNFRSF21:1288 TSLP:2022 LEF1:2815
Regulation Of Neuron Projection Developm	−0.06774229	165	2.752e−03	9.205e−01	MINAR1:7 CAMK2B:66 KANK1:117 ITGA6:199 ROR2:308 TENM3:434
Regulation Of Transcription By RNA Polym	0.20191780	22	1.046e−03	9.205e−01	CHD8:27 DDX21:145 ERCC6:204 DHX36:258 SMARCA5:399 ELL:471
Serine Family Amino Acid Biosynthetic Pr	0.27122738	9	4.839e−03	9.205e−01	CTH:608 SERINC3:745 MTHFD1:847 SRR:1930 PSPH:2220 SERINC5:2248
snRNA Transcription By RNA Polymerase II	0.31382881	7	4.036e−03	9.205e−01	SNAPC1:277 ELL:471 ZC3H8:725 ICE2:762 SNAPC5:2371 ELL3:8068
snRNA Transcription By RNA Polymerase II	0.43514501	4	2.577e−03	9.205e−01	SNAPC1:277 ZC3H8:725 ICE2:762 SNAPC5:2371 NA NA
Steroid Hormone Biosynthetic Process (GO	0.21240953	16	3.269e−03	9.205e−01	HSD17B6:146 CYP17A1:593 HSD17B3:600 HSD17B2:613 HSD17B11:893 CYP11A1:1087
Steroid Metabolic Process (GO:0008202)	0.11954974	61	1.254e−03	9.205e−01	HSD17B6:146 CYP51A1:314 HSD11B1:413 CYP17A1:593 HSD17B3:600 HSD17B2:613
Cilium Movement Involved In Cell Motilit	−0.30353125	7	5.419e−03	9.450e−01	TEKT1:157 TEKT2:1023 GAS8:1325 TEKT4:1573 TEKT5:2367 RSPH9:7900
Mitotic G1 DNA Damage Checkpoint Signali	−0.15646649	26	5.767e−03	9.450e−01	SDE2:264 MUC1:377 GML:389 PML:593 PLK3:753 GTSE1:952
Negative Regulation Of Endothelial Cell	0.15948978	25	5.788e−03	9.450e−01	NFE2L2:160 KRIT1:942 IL4:952 ICAM1:1194 FGG:1294 TEK:1440
Neutral Amino Acid Transport (GO:0015804	0.13712107	34	5.678e−03	9.450e−01	SLC43A1:32 SLC6A6:143 SLC7A6:157 SLC36A2:219 SLC36A1:338 SLC6A9:362
DNA Integrity Checkpoint Signaling (GO:0	0.11558151	36	1.645e−02	9.862e−01	TP53BP1:94 ERCC6:204 CDC45:603 PARP9:686 CLSPN:864 CDC6:885
DNA Replication Initiation (GO:0006270)	0.16029282	20	1.310e−02	9.862e−01	ORC1:65 ORC3:597 CDC45:603 ORC2:1104 LRWD1:1396 PRIM2:1588
L−phenylalanine Catabolic Process (GO:00	0.33126306	5	1.031e−02	9.862e−01	TAT:592 HGD:715 QDPR:1714 IL4I1:2352 GSTZ1:8068 NA
L−phenylalanine Metabolic Process (GO:00	0.33126306	5	1.031e−02	9.862e−01	TAT:592 HGD:715 QDPR:1714 IL4I1:2352 GSTZ1:8068 NA
L−serine Metabolic Process (GO:0006563)	0.24584037	10	7.106e−03	9.862e−01	SDS:598 SDSL:604 SERINC3:745 CBS:1692 SRR:1930 PSPH:2220
Anterograde Trans−Synaptic Signaling (GO	−0.05418828	180	1.242e−02	9.862e−01	MINK1:56 CHRNB2:60 HTR3B:177 CHRM3:241 CHRNA3:347 CHRNB1:383
Aromatic Compound Catabolic Process (GO:	0.26068214	7	1.692e−02	9.862e−01	CYP2F1:30 TET2:179 PON3:900 PDXP:1354 PON1:8068 PON2:8068

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	−0.17013844	127	3.725e−11	2.417e−07	LAMA5:61 HELZ2:64 NPBWR2:75 BIRC7:136 GATA5:200 ABHD16B:224
HAMAI_APOPTOSIS_VIA_TRAIL_UP	0.07614772	591	3.180e−10	1.032e−06	TASOR:14 SPG11:25 SMCHD1:35 CEP192:48 KIF14:54 NEK1:64
DACOSTA_UV_RESPONSE_VIA_ERCC3_DN	0.06143360	809	3.778e−09	8.171e−06	TMEM131L:6 HIRA:7 MAN2A1:13 TASOR:14 UBR2:18 SPG11:25
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	−0.28843801	33	9.833e−09	1.595e−05	DCST1:51 ADAM15:99 MTX1:134 PBXIP1:142 RUSC1:181 ZBTB7B:227
DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN	0.07693100	441	3.458e−08	4.487e−05	HIRA:7 MAN2A1:13 TASOR:14 UBR2:18 SMCHD1:35 ZNF292:40
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	−0.21378117	46	5.299e−07	5.730e−04	PIEZO1:11 CPNE7:192 ZC3H18:229 ANKRD11:334 PABPN1L:456 CBFA2T3:522
NIKOLSKY_BREAST_CANCER_21Q22_AMPLICON	−0.37576264	14	1.127e−06	1.044e−03	COL6A1:73 COL6A2:87 SPATC1L:165 PCNT:236 DIP2A:606 COL18A1:808
MARTENS_TRETINOIN_RESPONSE_UP	−0.05423703	675	1.803e−06	1.462e−03	B3GNT6:19 SMTNL2:29 MMEL1:30 CARD14:41 C1QTNF8:43 COL6A1:73
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	−0.34442946	15	3.863e−06	2.703e−03	SFSWAP:17 GALNT9:616 NOC4L:631 PUS1:636 GOLGA3:676 EP400:682
REACTOME_SIGNALING_BY_GPCR	−0.05401748	623	4.737e−06	2.703e−03	TACR2:22 GPR132:47 CAMK2B:66 NPBWR2:75 BDKRB1:86 ARHGEF11:147
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	−0.07733324	296	4.999e−06	2.703e−03	TACR2:22 GPR132:47 NPBWR2:75 BDKRB1:86 GPR68:150 CHRM3:241
DAZARD_RESPONSE_TO_UV_NHEK_DN	0.07986617	280	4.485e−06	2.703e−03	TMEM131L:6 WDFY3:37 KIF14:54 AKAP9:82 DOP1A:95 DST:106
WP_GPCRS_CLASS_A_RHODOPSINLIKE	−0.08442194	211	2.453e−05	1.224e−02	NPBWR2:75 BDKRB1:86 GPR68:150 OR6B1:207 CHRM3:241 SSTR4:256
STARK_HYPPOCAMPUS_22Q11_DELETION_DN	0.25685484	22	3.042e−05	1.410e−02	HIRA:7 PRODH:234 TANGO2:283 DGCR6:343 DGCR2:538 COMT:948
HSIAO_LIVER_SPECIFIC_GENES	0.08490669	199	3.749e−05	1.621e−02	GSTO1:20 F11:57 SERPIND1:67 ALDH1A1:85 ALB:142 MGST1:216
PEREZ_TP63_TARGETS	−0.06475642	313	8.548e−05	3.262e−02	CEMIP:6 NLRP1:49 KLK15:67 SEMA4D:155 APC2:212 EVPL:226
SCHLOSSER_SERUM_RESPONSE_DN	0.04662107	618	8.344e−05	3.262e−02	TASOR:14 UBR2:18 HEXB:21 ICAM2:28 ASMTL:29 PTPRC:47
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.10797444	101	1.792e−04	6.459e−02	SLC43A1:32 SLC6A6:143 SLC7A6:157 SLC36A2:219 SLC26A3:273 SLC36A1:338
PYEON_CANCER_HEAD_AND_NECK_VS_CERVICAL_U	0.08397830	166	1.930e−04	6.591e−02	NR1D2:91 WDR76:134 SLF1:196 NUP210:202 HLTF:206 TUG1:243
WP_2Q112_COPY_NUMBER_VARIATION_SYNDROME	0.18366797	34	2.108e−04	6.837e−02	ARID5A:98 CNNM4:178 MMS19:250 CNNM3:470 KANSL3:474 COL2A1:503
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	0.04466657	586	2.384e−04	7.366e−02	KIF14:54 WDR76:134 APC:147 AGGF1:167 SLF1:196 HLTF:206
REACTOME_GPCR_LIGAND_BINDING	−0.05360592	398	2.553e−04	7.370e−02	TACR2:22 GPR132:47 NPBWR2:75 BDKRB1:86 GPR68:150 CHRM3:241
DUTERTRE_ESTRADIOL_RESPONSE_24HR_UP	0.06065809	308	2.613e−04	7.370e−02	DEPTOR:88 WDR76:134 TET2:179 IMPA2:208 SPDL1:227 KCNK15:242
ZHANG_BREAST_CANCER_PROGENITORS_UP	0.05210130	408	3.214e−04	8.687e−02	PROM1:1 SMCHD1:35 CEP192:48 ANAPC1:62 NEK1:64 AKAP9:82
WP_PEPTIDE_GPCRS	−0.12205469	70	4.165e−04	1.055e−01	TACR2:22 BDKRB1:86 SSTR4:256 OPRK1:396 GALR1:440 FSHR:490
WP_22Q112_COPY_NUMBER_VARIATION_SYNDROME	0.10012894	104	4.230e−04	1.055e−01	HIRA:7 CLTCL1:19 PI4KA:36 SERPIND1:67 SLC7A4:120 DEPDC5:136
DODD_NASOPHARYNGEAL_CARCINOMA_DN	0.03016670	1204	4.957e−04	1.191e−01	GSTO1:20 CEP192:48 KIF14:54 ANAPC1:62 EMILIN2:66 QSER1:127
FISCHER_DREAM_TARGETS	0.03346534	865	9.215e−04	2.135e−01	MTBP:5 SMCHD1:35 GATAD2A:43 FNBP4:44 CEP192:48 KIF14:54
REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTOR	−0.07389589	168	9.673e−04	2.164e−01	TACR2:22 NPBWR2:75 BDKRB1:86 SSTR4:256 C3:296 NPW:340
ASGHARZADEH_NEUROBLASTOMA_POOR_SURVIVAL_	−0.15004354	39	1.188e−03	2.569e−01	ADCY1:495 C1orf35:576 GATAD2B:775 HAP1:901 H2AZ2:999 TMOD2:1054
CAIRO_HEPATOBLASTOMA_DN	0.06121765	234	1.285e−03	2.688e−01	TMEM131L:6 F11:57 OGDHL:93 CA2:140 HSD17B6:146 LEPR:190
BENPORATH_ES_WITH_H3K27ME3	−0.03042522	976	1.434e−03	2.745e−01	CEMIP:6 PLEC:12 BNC1:28 ANKRD35:31 CAMK2B:66 LONRF3:85
DAZARD_UV_RESPONSE_CLUSTER_G6	0.07923190	137	1.382e−03	2.745e−01	KIF14:54 DOP1A:95 TUT4:156 UBR5:166 ZNF451:187 OTUD4:201
TOYOTA_TARGETS_OF_MIR34B_AND_MIR34C	0.04654378	401	1.439e−03	2.745e−01	PPP4R1:10 KIF14:54 FAR2:118 DDX21:145 NUP210:202 RAP1GAP:232
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.36837516	6	1.778e−03	3.105e−01	CA2:140 CA4:195 SLC4A1:1012 RHAG:1018 CA1:2187 AQP1:8068
DEURIG_T_CELL_PROLYMPHOCYTIC_LEUKEMIA_DN	0.05567390	266	1.818e−03	3.105e−01	MAN2A1:13 TASOR:14 ABCB1:15 ZNF292:40 PTPRC:47 PJA2:71
KINSEY_TARGETS_OF_EWSR1_FLII_FUSION_UP	0.02751912	1172	1.699e−03	3.105e−01	PROM1:1 KIF14:54 ORC1:65 RADX:81 WDR76:134 DDX21:145
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.17034477	28	1.811e−03	3.105e−01	HSD17B6:146 HSD11B1:413 AKR1D1:460 CYP17A1:593 HSD17B3:600 HSD17B2:613
MEISSNER_NPC_HCP_WITH_H3K27ME3	−0.10326135	76	1.866e−03	3.105e−01	GATA5:200 B4GALNT2:455 SFRP5:479 TCFL5:770 GRM7:827 HCK:906
NIKOLSKY_BREAST_CANCER_16P13_AMPLICON	−0.08999657	99	1.988e−03	3.225e−01	C1QTNF8:43 EME2:286 NPW:340 UNKL:605 SYNGR3:738 NME3:781

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hypocalcemia	0.20366287	45	2.311e−06	2.267e−02	HIRA:7 CASR:63 IFT122:154 UBR1:214 DGCR6:343 CLDN16:423
DiGeorge Syndrome	0.15568988	61	2.646e−05	9.254e−02	HIRA:7 SPECC1L:138 ALB:142 PRODH:234 TJP1:282 DGCR6:343
Nephrocalcinosis	0.16482634	54	2.830e−05	9.254e−02	CLCNKB:41 NSD1:42 CASR:63 SLC26A3:273 CLDN16:423 SLC12A1:431
Tetany	0.25943809	21	3.874e−05	9.500e−02	HIRA:7 CASR:63 CLDN16:423 SLC12A1:431 TRPM6:624 GCM2:629
Shprintzen syndrome	0.20149727	33	6.218e−05	1.220e−01	HIRA:7 PRODH:234 DGCR6:343 DGCR6L:366 CHRNA7:511 DGCR2:538
Specific learning disability	0.12207327	86	9.300e−05	1.521e−01	HIRA:7 SPG11:25 NSD1:42 SIL1:165 POMK:279 DGCR6:343
22q11 partial monosomy syndrome	0.33633222	11	1.123e−04	1.574e−01	HIRA:7 PI4KA:36 JMJD1C:780 COMT:948 GNB1L:1306 ARVCF:1586
Pain, Postoperative	0.22719548	21	3.140e−04	3.850e−01	ABCB1:15 TNF:259 TRPV1:263 SCN11A:367 AP3B1:644 PTGS2:721
Anemia, hereditary spherocytic hemolytic	0.42171455	5	1.091e−03	6.798e−01	ANK1:89 EPB42:568 SLC4A1:1012 SPTB:1657 SPTA1:2807 NA
Acanthocytosis	0.30612581	10	8.023e−04	6.798e−01	CAD:346 KEL:409 VPS13A:420 XK:824 PIK3C2A:916 MTTP:1120
Cervical Squamous Intraepithelial Neopla	0.14217909	43	1.264e−03	6.798e−01	APC:147 TNF:259 MAL:315 FBXW7:525 MET:558 LAMA3:617
Idiopathic hypercalciuria	0.35501167	7	1.143e−03	6.798e−01	CASR:63 SLC12A1:431 VDR:964 SLC12A3:1299 CLCN5:2330 TRPV5:2880
Increased calcium level in kidney	0.16711728	33	8.965e−04	6.798e−01	NSD1:42 CASR:63 CLDN16:423 SLC12A1:431 COL4A3:456 GCM2:629
Occipital myelomeningocele	0.30946896	9	1.305e−03	6.798e−01	HIRA:7 JMJD1C:780 COMT:948 ARVCF:1586 TBX1:2367 RREB1:2560
Polyglandular Type I Autoimmune Syndrome	0.17870421	30	7.078e−04	6.798e−01	VCAM1:24 CASR:63 TNF:259 NLRP5:429 IL7:826 BPIFB1:1083
Seborrheic dermatitis	0.20752574	20	1.317e−03	6.798e−01	HIRA:7 NFE2L2:160 TNF:259 JMJD1C:780 BTD:820 COMT:948
Spherocytosis	0.42171455	5	1.091e−03	6.798e−01	ANK1:89 EPB42:568 SLC4A1:1012 SPTB:1657 SPTA1:2807 NA
Steroid−sensitive nephrotic syndrome	0.27741995	12	8.768e−04	6.798e−01	TNF:259 PLA2G7:432 HPSE:534 IL12B:1054 VEGFA:1244 GLO1:1472
Unilateral primary pulmonary dysgenesis	0.42529439	5	9.893e−04	6.798e−01	DGCR6:343 DGCR2:538 DGCR8:1230 ESS2:1363 TBX1:2367 NA
Congenital dyserythropoietic anemia, typ	0.25648679	12	2.096e−03	7.139e−01	MAN2A1:13 SEC23B:305 MCFD2:675 SLC4A1:1012 AICDA:1243 KLF1:1307
Mental Retardation, X−Linked 1	−0.13599417	42	2.304e−03	7.139e−01	DMD:194 ARHGEF6:359 AFF2:442 ZNF711:465 CLCN4:489 MED12:1355
PSEUDOHYPOALDOSTERONISM, TYPE IID	0.31334904	8	2.147e−03	7.139e−01	NEK1:64 PRPF4B:83 WNK1:934 SLC12A3:1299 WNK4:2264 TRPV5:2880
CONOTRUNCAL ANOMALY FACE SYNDROME	0.19196592	21	2.329e−03	7.139e−01	HIRA:7 DGCR6:343 DGCR2:538 JMJD1C:780 COMT:948 DGCR8:1230
Hepatitis	0.05696560	243	2.315e−03	7.139e−01	PROM1:1 ABCB1:15 VCAM1:24 ALB:142 APC:147 AFP:161
Hooked clavicle	0.36518280	6	1.949e−03	7.139e−01	NEK1:64 ORC1:65 INPPL1:452 SCARF2:1417 SLC35D1:2740 SKI:8068
Hyperactive renin−angiotensin system	0.31505693	8	2.030e−03	7.139e−01	CLCNKB:41 SLC26A3:273 SLC12A1:431 NR3C2:1125 SCNN1A:2564 SCNN1B:2892
Hypophosphatemia	0.13350809	46	1.742e−03	7.139e−01	CASR:63 MEPE:280 SLC34A3:433 COL4A3:456 ACOT8:506 GCM2:629
Impaired T cell function	0.21413780	17	2.241e−03	7.139e−01	HIRA:7 CR2:698 JMJD1C:780 UMPS:915 COMT:948 ARVCF:1586
Increased plasma renin activity	0.31505693	8	2.030e−03	7.139e−01	CLCNKB:41 SLC26A3:273 SLC12A1:431 NR3C2:1125 SCNN1A:2564 SCNN1B:2892
Malabsorption Syndrome	0.15291380	35	1.752e−03	7.139e−01	LCT:107 CNNM4:178 PPARA:373 PYY:462 NR1H4:570 AMN:750
Posterior embryotoxon	0.16367060	31	1.617e−03	7.139e−01	HIRA:7 DGCR6:343 DGCR2:538 JMJD1C:780 PEX13:891 COMT:948
Small earlobe	0.24674603	13	2.069e−03	7.139e−01	HIRA:7 JMJD1C:780 CDC6:885 COMT:948 ARVCF:1586 TBX1:2367
Color Blindness, Red	−0.35148375	5	6.490e−03	7.328e−01	TEX28:196 ATF6:799 PDE6H:1241 GNAT2:1803 OPN1LW:7900 NA
Leishmaniasis, New World	0.26207179	9	6.480e−03	7.328e−01	TNF:259 IL4:952 IFNG:1099 CBS:1692 CBSL:2353 IL10:2602
Malaria, Cerebral	0.09836754	69	4.762e−03	7.328e−01	VCAM1:24 BTNL2:53 APC:147 IL33:188 TNF:259 IFNAR1:351
PEROXISOME BIOGENESIS DISORDER, COMPLEME	0.30814788	7	4.753e−03	7.328e−01	MCCC1:476 BMS1:2241 TBX1:2367 PEX26:2427 ACACA:2726 CDC73:2965
22q11 Deletion Syndrome	0.19143268	19	3.873e−03	7.328e−01	HIRA:7 PRODH:234 HIC2:464 JMJD1C:780 COMT:948 DGCR8:1230
Aarskog syndrome	−0.23013379	12	5.777e−03	7.328e−01	ARHGEF2:221 ROR2:308 SLC2A4RG:561 FGD1:635 GLI3:1205 RASGRF1:1349
Abdominal wall muscle weakness	0.29517079	8	3.840e−03	7.328e−01	SMCHD1:35 VDR:964 MAP2K2:1819 NF1:2188 CYP27B1:2337 GYG1:2496
Abnormal lung lobation	−0.25772570	11	3.080e−03	7.328e−01	LBR:1034 AKT1:1126 GLI3:1205 GPC3:1702 DHCR7:1706 PTEN:1998

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0002136 abnormal kidney physiology	0.070211398	278	7.901e−05	3.761e−02	MAN2A1:13 CLCNKA:38 PTPRC:47 CASR:63 SLC14A2:78 ALB:142
MP0009643 abnormal urine homeostasis	0.066551734	226	6.964e−04	1.658e−01	MAN2A1:13 CLCNKA:38 CASR:63 SLC14A2:78 ALB:142 LMX1B:150
MP0005636 abnormal mineral homeostasis	0.081601070	134	1.236e−03	1.962e−01	CASR:63 ANK1:89 ALB:142 NFE2L2:160 SLC26A3:273 CLDN16:423
MP0001765 abnormal ion homeostasis	0.087814695	103	2.236e−03	2.661e−01	CASR:63 SLC26A3:273 CLDN16:423 SLC12A1:431 SLC34A3:433 TRPM6:624
MP0000678 abnormal parathyroid gland	0.188685527	21	2.798e−03	2.664e−01	CASR:63 GCM2:629 VDR:964 VEGFA:1244 ECE1:1878 FOXG1:2094
MP0005332 abnormal amino acid	0.075127892	116	5.572e−03	4.397e−01	SLC6A6:143 LEPR:190 PRODH:234 SLC12A1:431 COL4A3:456 BCAT2:565
MP0008469 abnormal protein level	0.041652930	384	6.466e−03	4.397e−01	HEXB:21 NFE2L2:160 LEPR:190 UBR1:214 PLG:255 TNF:259
MP0003656 abnormal erythrocyte physiolo	0.165177574	22	7.406e−03	4.407e−01	MAN2A1:13 NEK1:64 ANK1:89 NFE2L2:160 AQP9:994 SLC4A1:1012
MP0005319 abnormal enzyme/ coenzyme	0.059306196	170	8.409e−03	4.447e−01	NFE2L2:160 SLC12A1:431 NR1H4:570 MTMR3:573 TFAM:601 CTH:608
MP0002139 abnormal hepatobiliary system	0.261875589	8	1.036e−02	4.827e−01	ALDH1A1:85 FABP1:1092 INSIG1:1750 SLC22A1:1920 ABCG8:2239 NR1H3:8068
MP0010386 abnormal urinary bladder	−0.194443696	14	1.185e−02	4.827e−01	CHRNB2:60 CHRM3:241 CHRNA3:347 CHRNB4:1074 TAC1:1273 UPK2:1631
MP0001663 abnormal digestive system	0.078692478	86	1.217e−02	4.827e−01	APC:147 LEPR:190 COL19A1:245 SLC26A3:273 GLG1:370 COL2A1:503
MP0000598 abnormal liver morphology	0.033035345	466	1.831e−02	6.038e−01	MAN2A1:13 HEXB:21 F11:57 DNMT3B:58 ANK1:89 MANBA:112
MP0002837 dystrophic cardiac calcinosis	−0.226579521	9	1.865e−02	6.038e−01	SLC2A4:240 MMP9:1347 DNM1L:1374 DES:2226 SCG5:2417 GSTZ1:7900
MP0002089 abnormal postnatal growth/wei	0.128280849	28	1.903e−02	6.038e−01	COL2A1:503 MMP14:881 OCA2:1468 MYC:1794 BAX:2009 LAMA2:2046
MP0005334 abnormal fat pad	0.051611830	149	3.145e−02	8.195e−01	PPP1R3A:137 LEPR:190 ERCC6:204 ATXN2:224 TRPV1:263 HDC:365
MP0005381 digestive/alimentary phenotyp	0.059105910	109	3.440e−02	8.195e−01	APC:147 LEPR:190 COL19A1:245 SLC26A3:273 GLG1:370 COL2A1:503
MP0002135 abnormal kidney morphology	0.030618210	448	3.171e−02	8.195e−01	LRP4:11 MAN2A1:13 HEXB:21 PTPRC:47 NEK1:64 ANK1:89
MP0000313 abnormal cell death	0.029929906	449	3.556e−02	8.195e−01	PROM1:1 HEXB:21 LAMA1:33 DNMT3B:58 DAXX:80 APC:147
MP0001756 abnormal urination	0.068708198	80	3.464e−02	8.195e−01	CLCNKA:38 PTPRC:47 SLC14A2:78 DST:106 ALB:142 LEPR:190
MP0001873 stomach inflammation	0.132257136	21	3.616e−02	8.195e−01	IL4:952 TOP3B:1010 SLC9A4:1056 IFNG:1099 OLFM4:1806 RELB:1996
MP0000230 abnormal systemic arterial	0.050902843	141	3.886e−02	8.408e−01	LEPR:190 RORA:257 PPARA:373 SLC12A1:431 COL4A3:456 ABCC9:512
MP0003195 calcinosis	0.207021678	8	4.270e−02	8.690e−01	CASR:63 SLC12A1:431 AGXT:650 SPP1:2441 ABCC6:2717 SOD2:8068
MP0000343 altered response to	−0.081104791	52	4.381e−02	8.690e−01	SLC2A4:240 MMP7:314 TGM2:316 CYSLTR2:353 CCR5:592 NPR1:839
MP0005000 abnormal immune tolerance	0.037988399	239	4.682e−02	8.914e−01	MAN2A1:13 PTPRC:47 NFE2L2:160 TNFSF15:164 LEPR:190 TNF:259
MP0000609 abnormal liver physiology	0.030422492	370	5.057e−02	8.915e−01	MAN2A1:13 PTPRC:47 ALB:142 NFE2L2:160 LEPR:190 ERCC6:204
MP0005395 other phenotype	0.061534252	86	4.993e−02	8.915e−01	CASR:63 DST:106 SLC23A1:405 NME1:424 NLRP5:429 SLC12A1:431
MP0004043 abnormal pH regulation	0.144299464	14	6.180e−02	9.524e−01	SLC12A1:431 SLC26A7:625 SLC4A7:1933 WNK4:2264 SCNN1B:2892 HIF1AN:8068
MP0000955 abnormal spinal cord	0.037847810	211	6.203e−02	9.524e−01	HEXB:21 CUX2:68 MANBA:112 LTN1:135 LEPR:190 INTU:246
MP0004185 abnormal adipocyte glucose	0.141877438	15	5.735e−02	9.524e−01	PPARG:1140 PRKCA:1506 E2F1:2034 TNKS2:2384 NPY1R:2479 PRKCQ:2609
MP0003634 abnormal glial cell	0.038574995	203	6.190e−02	9.524e−01	CRB1:26 LAMA1:33 PTPRC:47 LTN1:135 NFE2L2:160 LRRK2:240
MP0003699 abnormal female reproductive	0.027028543	411	6.829e−02	9.872e−01	UBR2:18 NEK1:64 SLC6A6:143 AFP:161 LEPR:190 DAZL:203
MP0002882 abnormal neuron morphology	0.019475919	873	6.844e−02	9.872e−01	PROM1:1 LRP4:11 HEXB:21 CRB1:26 CUX2:68 ANK1:89
MP0008438 abnormal cutaneous collagen	−0.081439537	15	2.753e−01	9.992e−01	OGN:600 DSE:1421 PLOD1:1680 DCN:2648 TNXB:7900 PLOD3:7900
MP0002909 abnormal adrenal gland	−0.006838714	21	9.137e−01	9.992e−01	SMPD3:1200 ITSN1:1715 APOE:1751 SLC23A2:2782 FOXA1:7900 SNAPIN:7900
MP0001730 embryonic growth arrest	0.022578146	179	3.037e−01	9.992e−01	HIRA:7 CHD8:27 GATAD2A:43 UGDH:101 UBR1:214 PTPRB:307
MP0001731 abnormal postnatal growth	−0.007860789	585	5.344e−01	9.992e−01	PLEC:12 CHRNB2:60 TNS3:119 AP3D1:148 AEBP1:153 GRIN3B:167
MP0002210 abnormal sex determination	0.013019555	325	4.309e−01	9.992e−01	NEK1:64 EIF4G3:72 TAF7L:141 DAZL:203 TNF:259 PIWIL2:297
MP0000604 amyloidosis	−0.050671488	33	3.148e−01	9.992e−01	APCS:154 BACE1:508 CCR2:1088 DLST:1303 CTSK:1554 APOE:1751
MP0005330 cardiomyopathy	0.015760108	68	6.547e−01	9.992e−01	SLC6A6:143 PPARA:373 ABCA5:546 TFAM:601 PLRG1:1221 SLC6A4:2391

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
pancreas	0.094779632	94	1.567e−03	6.155e−02	SLC43A1:32 F11:57 CASR:63 CELA3A:113 PRODH2:128 CELA3B:177
thyroid	0.072693027	150	2.280e−03	6.155e−02	HIRA:7 CLCNKA:38 CLUL1:60 RAP1GAP:232 GPAT3:262 TMC4:376
liver	0.045218250	327	5.668e−03	1.020e−01	MOGAT1:16 F11:57 SERPIND1:67 PRODH2:128 ALB:142 HSD17B6:146
EBV.lymphocyte	0.030726593	669	8.791e−03	1.187e−01	MTBP:5 TMEM131L:6 KIF14:54 ORC1:65 WDR76:134 NUP210:202
esophagus.mucosa	−0.042797254	220	3.043e−02	3.286e−01	IQANK1:8 BNC1:28 BDKRB1:86 RNF222:112 FAM83H:197 CDKN2B:215
cerebral.cortex	−0.055354935	107	4.902e−02	4.412e−01	CHRNB2:60 CALHM1:98 HTR3B:177 ADGRA1:208 KCNS1:284 RCVRN:470
vagina	−0.051936327	90	8.992e−02	6.936e−01	BNC1:28 RNF222:112 EVPL:226 SLC18A2:403 SLURP1:412 ENDOU:567
amygdala	−0.134019166	9	1.640e−01	7.383e−01	PCDH15:203 SLC38A8:1196 NEUROD6:1860 SLC17A7:2054 BCAN:7900 LHFPL3:7900
cortex.kidney	0.029852919	114	2.734e−01	7.383e−01	CLCNKB:41 CASR:63 OGDHL:93 PRODH2:128 SLC36A2:219 CLDN16:423
ectocervix	0.068231638	25	2.381e−01	7.383e−01	PAPPA2:478 LRFN5:918 RIPOR3:1255 MMRN1:1745 ADRA1D:1861 KRT34:2650
esophagus.muscularis.mucosa	−0.069571973	26	2.200e−01	7.383e−01	SYNM:88 PRUNE2:145 CHRM3:241 COL4A6:385 FAM83D:967 KCNMB1:1352
hypothalamus	−0.055691881	66	1.188e−01	7.383e−01	PCDH15:203 DIRAS3:326 LY6H:375 DLX2:723 SYNGR3:738 GRM7:827
left.ventricle	−0.048416354	68	1.687e−01	7.383e−01	MB:144 FSD2:149 MYH7B:235 ASB11:765 TNNT2:1364 ADPRHL1:1550
leg.skin	−0.023064222	217	2.466e−01	7.383e−01	BNC1:28 ANKRD35:31 CARD14:41 POF1B:83 FAM83G:101 C2CD4D:105
nucleus.accumbens	−0.039276580	99	1.789e−01	7.383e−01	PCDH11X:77 DACH2:232 OPRK1:396 OTOF:535 KIRREL3:722 NTSR2:743
prostate	−0.051659678	38	2.713e−01	7.383e−01	ANO7:54 CREB3L4:71 HOXB13:282 CDH26:538 LMAN1L:540 ACP3:829
sigmoid.colon	−0.056235294	50	1.699e−01	7.383e−01	TACR2:22 PLA2G2C:173 ROR2:308 CHRNA3:347 NTSR1:833 PDE9A:1107
skeletal.muscle	−0.021520535	238	2.582e−01	7.383e−01	SMTNL2:29 ALPK3:58 TACC2:95 MB:144 FSD2:149 RGS9BP:183
substantia.nigra	0.097666714	12	2.416e−01	7.383e−01	DBX2:1455 RET:2498 CHRM5:2518 FOXB1:8068 C10orf105:8068 KCNE5:8068
testis	−0.012868539	1375	1.377e−01	7.383e−01	DNAH14:2 PKD1L1:5 DNAH2:9 FBXW10:13 ZBED9:15 POM121L2:16
breast	−0.072129760	15	3.337e−01	8.582e−01	THRSP:318 ABCB5:702 TNN:1590 ALX4:2081 ADIPOQ:7900 CIDEC:7900
bladder	0.050345996	24	3.937e−01	8.632e−01	CYP4B1:1226 UPK1A:1487 SLC14A1:1601 UPK3A:2243 PLA2G4A:2349 SNX31:2628
ovary	−0.024251511	102	3.996e−01	8.632e−01	MMP21:74 TCF23:82 SUSD4:143 KLHDC8A:510 TSHZ3:521 GLI2:645
suprapubic.skin	−0.018103981	193	3.902e−01	8.632e−01	BNC1:28 CARD14:41 RNF222:112 GGT6:131 EVPL:226 PNLIPRP3:271
eye.development	−0.030238831	67	3.935e−01	8.632e−01	TKT:210 DLX2:723 FGFR2:751 FOXC2:1100 BMP7:1494 NR2F2:1839
tibial.artery	−0.022980372	106	4.160e−01	8.640e−01	C1QTNF1:478 SUSD5:575 GPR20:609 CDH6:622 COL18A1:808 MCAM:909
atrium.auricle	−0.022458224	90	4.634e−01	8.728e−01	ADAMTS7:26 MB:144 FSD2:149 MYBPHL:211 LMAN1L:540 ASB11:765
coronary.artery	−0.052120357	15	4.849e−01	8.728e−01	SUSD5:575 TIMP1:1501 ST6GALNAC5:1776 CCN3:7900 SPINK13:7900 MYO18B:7900
endocervix	−0.026505996	72	4.383e−01	8.728e−01	BDKRB1:86 NXNL2:113 PWP2:124 ZNF516:171 TGM2:316 SLC18A2:403
hippocampus.proper	−0.065482734	10	4.735e−01	8.728e−01	NEUROD6:1860 SLC17A7:2054 OLIG2:2637 NEUROD2:7900 HRK:7900 FEZF2:7900
adrenal.gland	−0.010389373	108	7.105e−01	9.793e−01	SLC47A1:35 ZNF275:339 ABCC3:351 CYSLTR2:353 GML:389 KLHL4:560
aorta	0.006403825	154	7.854e−01	9.793e−01	ATP10A:12 ARID5A:98 KCNK15:242 LGR6:289 COL14A1:392 KCNK17:828
blood	0.004571260	274	7.972e−01	9.793e−01	PTPRC:47 STK10:108 SLC6A6:143 LY6G6F:173 GCA:210 TSPO2:251
Brodmann.area.24	−0.020844164	31	6.883e−01	9.793e−01	KCNS1:284 TAC3:1426 SLC39A12:1835 NEUROD6:1860 SLC17A7:2054 CSPG5:2388
Brodmann.area.9	−0.008742078	180	6.884e−01	9.793e−01	CHRNB2:60 CALHM1:98 HTR3B:177 NRCAM:204 DACH2:232 KCNS1:284
C1.spinal.cord	−0.009889792	182	6.483e−01	9.793e−01	ACSBG1:46 MOG:102 TMEM235:106 SEMA4D:155 FAM124A:338 PLD4:381
caudate.nucleus	0.003107784	26	9.563e−01	9.793e−01	FAM237A:919 MLC1:2186 GPR37L1:2638 ETNPPL:2937 HS3ST5:8068 ANO3:8068
cerebellar.hemisphere	0.007519455	500	5.744e−01	9.793e−01	MTBP:5 CRB1:26 ANK1:89 AFF3:99 CA4:195 TLL1:269
cerebellum	−0.003994607	374	7.945e−01	9.793e−01	GLRA4:42 SPHKAP:52 TRANK1:110 CNPY1:111 SLC22A31:179 APC2:212
esophagogastric.junction	−0.038590249	8	7.055e−01	9.793e−01	CHRM3:241 BARX1:2647 ADCY5:7900 COL4A5:7900 GADL1:7900 HOXA4:7900